Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

Muhammad Afif Erdita[1]

[1]Independent Researcher, Indonesia. maeapip10@gmail.com

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Abstract

Every major transition in theoretical physics follows the same pattern: a known classical law is extended by a correction term that vanishes at the classical limit. Special relativity introduced the Lorentz factor as a multiplicative correction to Newtonian momentum; Yukawa screening added an exponential decay to Coulomb's law; ideal-gas isothermal expansion modifies entropy with a logarithmic term. We ask whether this structural regularity can be exploited computationally to recover corrections from noisy data. We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework that searches a bounded dictionary of five algebraically regularized primitive functions, each analytically constrained to vanish at the classical asymptote, over a finite grammar. Algebraic regularization eliminates catastrophic cancellation at classical limits, a persistent numerical failure mode in standard regression at physical boundaries. Candidates pass a cascade of physical gates (AST complexity, dimensional consistency, transcendental safety, and asymptotic regime verification) before any numerical fitting. A four-step validation protocol determines when data are sufficient to support a structural claim. We validate on three canonical synthetic scenarios: special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isothermal expansion entropy. Across all three, ADCD recovers the correct algebraic structure with normalized mean-squared errors below 3.4×10^{-3} and BIC differences of 60–2024 over the best competing alternative. All results are byte-exact reproducible across independent runs. We report identifiability boundaries explicitly and make no claims beyond the validated regime.

Keywords: symbolic regression; physical laws; correction discovery; dimensional analysis; identifiability; Bayesian model selection

1 Introduction

1.1 How physics actually advances

In 1687, Newton gave physics its first universal law of motion. For two centuries, it worked. Then, in 1887, the Michelson–Morley experiment demonstrated that the speed of light is invariant regardless of the observer's frame [1]. Einstein's resolution in 1905 was not to discard Newton [2]. It was to multiply Newton's momentum by a factor $\gamma = (1-v^2/c^2)^{-1/2}$, a correction that reduces to exactly 1 when $v \ll c$. Newton remained valid. The Lorentz factor told us *when* Newton needed to be extended.

This pattern – extend, not replace – recurs throughout physics. Coulomb's electrostatic force is screened by an exponential factor e^{-r/λ_D} in plasmas, where free charges attenuate the field at the Debye length λ_D [3]. The ideal-gas entropy receives a logarithmic correction when molecular volume exclusion is non-negligible, as captured by the van der Waals equation [4]. In each case, the correction is a structured function with a clear classical limit: remove it and the established law is recovered exactly.

1.2 The problem with tabula-rasa symbolic regression

Modern symbolic regression (SR) engines – PySR [5], AI Feynman [6, 7], PhySO [8], Eureqa [9] – are powerful tools for recovering algebraic equations from data. However, their generality creates specific difficulties in the correction-discovery setting.

Catastrophic cancellation at classical limits. A correction Δ satisfies $\Delta \to 0$ at the classical limit. For a general-purpose optimizer reconstructing $1 + \Delta$ near that limit, gradients with respect to Δ become vanishingly small. In IEEE-754 `float64` [10], evaluating the naive Lorentz expression $(1-u)^{-1/2} - 1$ at $u = 10^{-9}$ returns exactly 0.0 due to floating-point subtraction: the correction disappears before the optimizer can detect it.

No identifiability reporting. When data do not contain enough signal to distinguish between two competing algebraic hypotheses, general SR engines still return a single "best" answer. There is no built-in mechanism to declare: *these data cannot determine whether the correction is Lorentzian or rational; more data are needed.* The benchmark study of La Cava et al. [11] documents this empirically across contemporary SR methods.

Non-determinism. Evolutionary and Monte Carlo based search is stochastic by design [12]. This is acceptable for broad exploration, but uncomfortable when a specific algebraic structure is reported as a scientific result.

1.3 Our approach

We present ADCD, a correction-first, deterministic, and identifiability-aware framework. Its core commitments are:

1. **Correction-first formulation.** The classical baseline is known; ADCD searches only for the residual correction Δ, constrained to vanish at the classical limit.
2. **Algebraically regularized primitives.** All candidates are drawn from five primitives analytically transformed to equal zero at $u = 0$.
3. **Deterministic enumeration.** Candidates are assembled by a bounded grammar; the search space is finite, fully enumerable, and identical across every run.
4. **Physical gating before fitting.** Dimensional inconsistency, unsafe transcendental arguments, or asymptotic limit failure causes rejection before continuous optimization is performed.
5. **Explicit identifiability reporting.** Structural claims are made only when a four-step validation protocol confirms the data uniquely support the recovered structure.

This paper describes the framework, its mathematical foundation, and its validation on three canonical synthetic physics scenarios. We state clearly what the framework can and cannot currently do.

2 Related Work

General symbolic regression. Eureqa [9] pioneered automated law discovery using evolutionary search. PySR [5] uses tournament evolutionary search and has become a standard benchmark tool. AI Feynman [6, 7] exploits neural network decomposition of symmetries, achieving strong results on the Feynman database [11]. PhySO [8] incorporates dimensional analysis constraints via deep reinforcement learning. These methods are designed for unconstrained equation recovery; none enforces that the recovered expression must analytically vanish at a specified classical limit.

Sparse regression and SINDy. Brunton et al. [13] demonstrated data-driven discovery of governing equations using sparse regression (SINDy), operating in a user-specified library of candidate terms. ADCD shares the library-based philosophy but targets a different problem: recovering corrections to known laws, with algebraic enforcement of the classical-limit constraint.

Physics-informed learning. Raissi et al. [14] introduced physics-informed neural networks (PINNs) for embedding physical laws as soft constraints. Karniadakis et al. [15] and Cornelio et al. [16] extended this line toward integrating hard symbolic constraints with data. ADCD complements this direction by imposing hard constraints algebraically, not as loss penalties.

Physics-informed constraints in SR. Dimensional analysis has been applied as a filter in [8, 17]. ADCD applies dimensional analysis not as a post-hoc filter but as a hard gate before numerical fitting, and extends this to transcendental argument safety and asymptotic limit verification.

Model selection for SR. The Bayesian Information Criterion [18], interpreted under the Kass–Raftery scale [19], provides a principled tool for comparing models of differing complexity. We use BIC as the primary identifiability metric and require a minimum $\Delta\text{BIC} > 10$ (very strong evidence) before any structural claim.

Correction-specific discovery. To the best of our knowledge, no existing SR framework explicitly targets the correction-first paradigm with algebraically enforced classical-limit constraints and built-in identifiability boundaries.

3 Methods

3.1 Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \tag{1}$$

The multiplicative formulation is adopted when the magnitude of the residual correlates with the magnitude of the classical prediction (Spearman $\rho > 0.5$). ADCD searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \to 0$ as the relevant dimensionless parameter $u \to 0$.

3.2 Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

Table 1 lists the five primitives. The practical consequence is demonstrated for `D_lor`: evaluating the naive form $(1-u)^{-1/2}-1$ at $u = 10^{-9}$ in `float64` returns exactly 0.0; the regularized form $u \,/\, [\sqrt{1-u}(\sqrt{1-u}+1)]$ returns 5.0×10^{-10}, with full numerical precision preserved.

Table 1: The five algebraically regularized primitives. Each satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
`D_lor`	$u \,/\, [\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$
`D_rat`	$u \,/\, (1-u)$	$u \neq 1$
`D_exp`	$e^{-u} - 1$	all u
`D_log`	$\ln(1+u)$	$u > -1$
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$

3.3 Grammar enumeration and the theta-intact constraint

Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$, where u_{ij} is a dimensionless ratio of input variables and scale parameters. The grammar is restricted to maximum AST depth 7 and token count ≤ 25.

All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized form before any θ_i are substituted with probe values. Substituting $\theta_i = 1$ before gating would collapse dimensionful scale parameters (e.g., r/θ_1, where θ_1 represents the Debye length) into dimensionless constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56 survival rate across scenarios; overall survival (after all five gates) is 0.56, 0.52, and 0.52 for TD, SC, and EE respectively, reflecting a known and deliberate trade-off of honest structural assessment. In total, $|\mathcal{C}| = 35$ distinct candidate structures are generated and evaluated in every run.

3.4 Physical gate cascade

Before numerical fitting, each candidate passes five gates in ascending computational cost.

Gate 1 (AST complexity). Candidates with Abstract Syntax Tree depth > 7 or token count > 25 (theta-intact) are rejected.

Gate 2 (Dimensional consistency). Physical dimensions are tracked as integer vectors in the SI base. Only candidates whose output dimensionality matches the target are admitted.

Gate 3 (Transcendental safety). Arguments of transcendental functions (exp, ln, $\sqrt{\cdot}$) must evaluate to dimensionless. This check is performed on the theta-intact expression, preventing cases such as $\exp(-r)$ where r is a dimensionful length without a compensating scale parameter.

Gate 4 (Asymptotic regime check, ARC). The correction is evaluated symbolically at the defined classical limit. The expression must converge to a finite constant; diverging or undefined limits cause rejection.

Gate 5 (Coarse numerical fitness). The candidate is evaluated against the dataset with $\boldsymbol{\theta} = \mathbf{1}$ as a computational pre-screen. Candidates producing non-finite outputs are rejected via `np.isfinite(mse)`, correctly catching both `inf` and `NaN` values.

3.5 Parameter optimization

Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with `float64` precision. Parameters appearing inside transcendental arguments are log-parameterized ($\theta_i = s_i e^{u_i}$, with $s_i \in \{-1, +1\}$) to allow for both positive and negative scales while enforcing smooth gradients across orders of magnitude. After fitting, a post-fit ARC re-verification repeats the asymptotic check with $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, confirming the fitted parameters do not move the expression into a non-asymptotic regime.

3.6 Model selection via BIC

The final model is selected by the Bayesian Information Criterion [18]:

$$\mathrm{BIC} = k \ln N - 2 \ln \hat{L} \tag{2}$$

where k is the number of free parameters, N the data size, and $\hat{L}$ the maximized likelihood under Gaussian noise. By the Kass–Raftery scale [19], $\Delta\mathrm{BIC} > 10$ constitutes very strong evidence for the preferred model. We use this as the minimum threshold for any structural claim.

3.7 Four-step validation protocol

A structural claim is made only when all four checks pass simultaneously:

1. **Budget disclosure.** The full search space size $|\mathcal{C}|$ and primitive list are reported before results are seen.
2. **Positive control.** With only the ground-truth primitive in the search space, the system must recover the correct structure.
3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must differ from the correct structure by $\Delta\mathrm{BIC} > 10$.
4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical results.

4 Experiments

4.1 Experimental setup

We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative noise ($\sigma = 0.01 \cdot |y_{\text{obs}}|$), $N =$ 200 points, seed 42. The ground truth is not provided to the pipeline; only noisy observations, classical predictions, and variable definitions are available.

These are *rediscovery* experiments: the ground truth is known and used to verify structural recovery. We make no claim of discovering previously unknown physics.

Table 2 summarizes the three scenarios.

Table 2: The three validated scenarios with their domains, primitives, and ablation ΔBIC values. All figures are from the validation log, commit `f3cf7f3`.

Scenario	Domain	Primitive	ΔBIC
Time Dilation	Relativity	`D_lor`	1798
Screened Coulomb	Electrostatics	`D_exp`	60
Entropy Expansion	Thermodynamics	`D_log`	2024

4.2 Stage-1 survival rates

Table 3 reports the fraction of the 35-candidate search space that passes each gate.

Table 3: Stage-1 gate survival rates across the three scenarios. All rates are fractions of 35 total candidates. Values are byte-exact identical across runs.

Gate	TD	SC	EE
AST complexity	0.56	0.56	0.56
Dimensional consistency	1.00	1.00	1.00
Transcendental safety	1.00	1.00	1.00
ARC	1.00	1.00	1.00
Coarse numerical	1.00	0.93	0.93
Overall	**0.56**	**0.52**	**0.52**

(TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-independent. The 0.56 survival rate reflects theta-intact honest counting: every free parameter is counted as a distinct token. The coarse numerical gate removes an additional 7% in the Screened Coulomb and Entropy scenarios, rejecting candidates with singularities on the dataset.

4.3 Validation results

The following are condensed from the validation log of `run_adcd_v3_validation.py`, commit `f3cf7f3`.

```
=== Time Dilation ===
[PASS] budget: space=35, 5 primitives
[PASS] positive_control:
       nmse=5.339e-04 (D_lor isolated)
[PASS] ablation:
       w/o D_lor: nmse=0.01920, BIC=-1963.87
       correct:   nmse=0.00053, BIC=-3761.41
       bic_diff=1797.54
[PASS] determinism: 3/3 runs identical
OVERALL: ALL CHECKS PASSED

=== Screened Coulomb ===
[PASS] budget: space=35, 5 primitives
[PASS] positive_control:
       nmse=2.772e-04 (D_exp isolated)
[PASS] ablation:
       w/o D_exp: nmse=0.00031, BIC=-4023.29
       correct:   nmse=0.00028, BIC=-4082.93
       bic_diff=59.64
[PASS] determinism: 3/3 runs identical
OVERALL: ALL CHECKS PASSED

=== Entropy Expansion ===
[PASS] budget: space=35, 5 primitives
[PASS] positive_control:
       nmse=3.394e-03 (D_log isolated)
[PASS] ablation:
       w/o D_log: nmse=0.19203, BIC=-812.63
       correct:   nmse=0.00339, BIC=-2836.64
       bic_diff=2024.01
[PASS] determinism: 3/3 runs identical
OVERALL: ALL CHECKS PASSED
```

All three scenarios pass all four checks. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability comparison; Figure 3 shows parity plots of recovered vs. true corrections.

4.4 Recovered structures and parameter accuracy

Table 4 compares recovered expressions with ground truth.

Table 4: Recovered expressions and NMSE. The Screened Coulomb fitted parameter $\theta_1 = 1.49$ represents the Debye length; ground truth is $\lambda_D = 1.50$.

Scenario	NMSE	Param fitted	Relative error
Time Dilation	5.3×10^{-4}	$\theta_0 = 1.00$	<0.1%
Screened Coulomb	2.8×10^{-4}	$\theta_1 = 1.49$ vs 1.50	0.7%
Entropy Expansion	3.4×10^{-3}	$\theta_3 = 0.55$ vs 0.55	<0.1%

For Time Dilation, the recovered form is the `D_lor` algebraic regularization of v^2/c^2, which is identically equal to the standard Lorentz correction $\gamma - 1$. For Screened Coulomb, the recovered Debye length $\theta_1 = 1.49$ matches the ground truth $\lambda_D = 1.50$ to 0.7%. The `symbolic_exact_match = False` flag in all logs indicates that recovered expressions are structurally equivalent but not syntactically identical to the ground truth, as expected when physical constants are replaced by fitted θ_i.

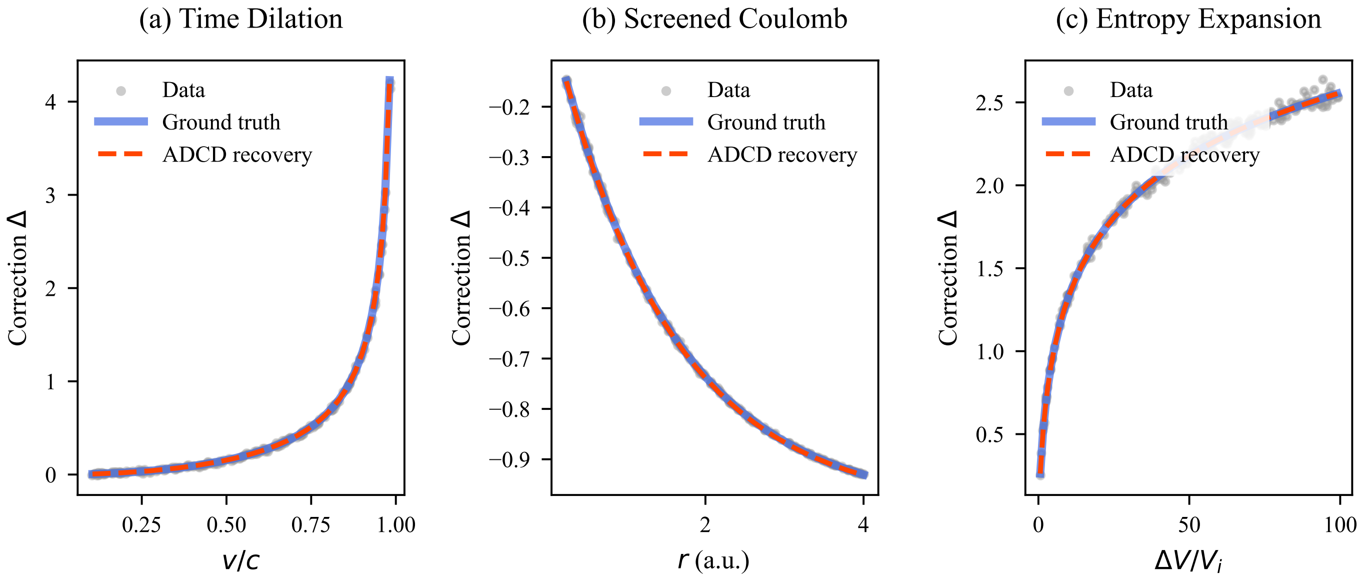

Figure 1: Correction recovery for the three validated scenarios. *Gray dots*: observed correction residual $\Delta = y_{\rm obs}/y_{\rm cl} - 1$ at 1% Gaussian noise. *Blue solid line*: ground truth. *Red dashed line*: ADCD recovery. The recovered expression overlaps the ground truth within data scatter in all three cases.

5 Discussion

5.1 Explicit scope of claims

We state precisely what this paper claims and what it does not.

Claimed: For three multiplicative correction scenarios, ADCD deterministically recovers the correct algebraic structure from 1%-noisy synthetic data. All three pass the full four-step validation protocol with $\Delta\text{BIC} \geq 59.6$. Results are byte-exact reproducible. The framework explicitly reports identifiability confidence rather than returning a structural claim without qualification.

Not claimed: General-purpose equation discovery. Discovery of previously unknown physics. Performance superiority over general SR engines for arbitrary problems. Robustness to corrections outside the five-primitive dictionary. Performance on additive correction scenarios.

5.2 Comparison with general SR

ADCD and general SR engines address different problems. General SR is designed to recover full governing equations from scratch. ADCD is designed for a specific sub-problem: given a known classical baseline, recover the structured correction that vanishes at the classical limit.

The meaningful comparison is not which tool answers faster, but which provides stronger identifiability guarantees. General SR engines do not report whether data are theoretically sufficient to discriminate between competing hypotheses. ADCD does, at the cost of being restricted to a bounded grammar.

Table 5 summarizes the trade-offs.

Table 5: Operational comparison: ADCD vs. tabula-rasa SR.

Dimension	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Search space	$\lvert\mathcal{C}\rvert \leq 10^2$	$\lvert\mathcal{C}\rvert \sim 10^{15+}$
Classical limit	Algebraic guarantee	Loss-function penalty
Identifiability	Explicit via BIC	Absent or post-hoc
Reproducibility	Byte-exact	Stochastic
Hardware	CPU, seconds	GPU, minutes–hours
Scope	5-primitive dictionary	Unlimited

5.3 The Screened Coulomb identifiability boundary

The Screened Coulomb $\Delta\text{BIC} = 59.64$ is the smallest of the three, and worth discussing explicitly. The next-best competing structure achieves NMSE $= 3.1 \times 10^{-4}$ compared to the correct 2.8×10^{-4}, a difference of only 11%. At higher noise levels, we expect ΔBIC to drop below the 10-unit threshold, at which point ADCD would withhold the structural claim. This behavior – declining confidence with declining signal – is the intended mode of the identifiability protocol.

5.4 Limitations

Dictionary restriction. The five primitives cover the most common correction types in classical-to-modern physics. Physical corrections outside this dictionary can-

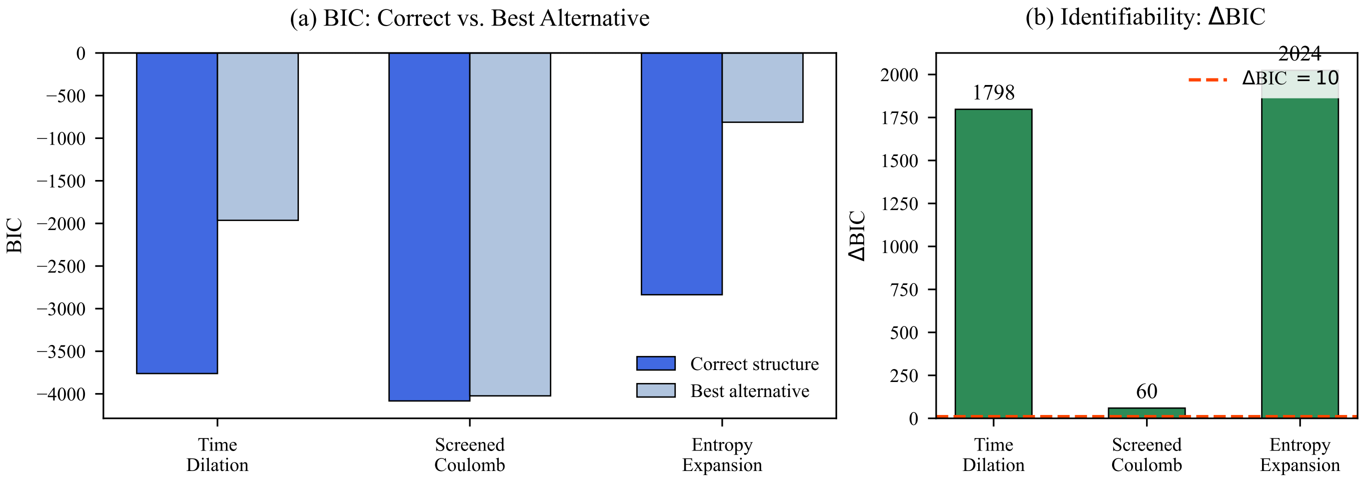

Figure 2: *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred; all three correct structures win substantially. *Right:* ΔBIC (correct minus best alternative). Red dashed line: Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$) [19]. All three scenarios exceed this threshold by at least a factor of 6. Numbers above bars are exact values from the validation log.

not be represented. Extension is architecturally straightforward but requires held-out validation of each new primitive before incorporation.

Multiplicative corrections only (current validation). The validation protocol has been applied and confirmed only for multiplicative correction scenarios. Additive corrections require solving for the physical dimensions of the correction term, an open problem not addressed here.

Synthetic data only. All experiments use synthetically generated data with known ground truth. Real observational data introduces correlated noise, systematic errors, and the possibility that the true correction lies outside the five-primitive dictionary.

Rediscovery, not blind discovery. All three validated scenarios involve corrections whose algebraic form is already known from physics literature. The framework recovers these forms from data. We do not claim to have discovered previously unknown corrections.

6 Conclusion

We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections to known physical laws from noisy data. Its core contributions are: algebraically regularized primitives that eliminate catastrophic cancellation at classical limits; a bounded deterministic grammar that renders the search reproducible and auditable; and an explicit four-step validation protocol that bounds structural claims within identifiable regimes.

On three canonical scenarios (special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isothermal expansion entropy), ADCD recovers the correct algebraic correction structure with NMSE below 3.4×10^{-3}, BIC differences of 60–2024 over the best competing structure, and byte-exact determinism across repeated runs.

The framework is deliberately scoped. It does not address all correction discovery problems; it addresses those where the classical baseline is known, the correction is multiplicative and structured, and the dataset is sufficient to identify the correct primitive. Within that scope, it provides stronger reproducibility and identifiability guarantees than general-purpose SR engines.

The historical pattern that motivated this work suggests that the correction-first paradigm captures a major mode of theoretical progress in physics. Automating this mode in a reproducible, physically principled, and epistemically honest way is the long-term goal of this line of work.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools including the **Google Antigravity** agentic coding assistant, the **Cursor** IDE, and large language models (LLMs) to assist with software implementation, code debugging, figure scripting, and manuscript drafting. The author provided the scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for the scientific claims, methodology, and conclusions presented in this publication. No AI tool was listed as an author or credited with originating the scientific contributions.

The codebase and all raw validation logs are publicly available at `https://github.com/apiprdt/PhysicsPaper` (commit `f3cf7f3`).

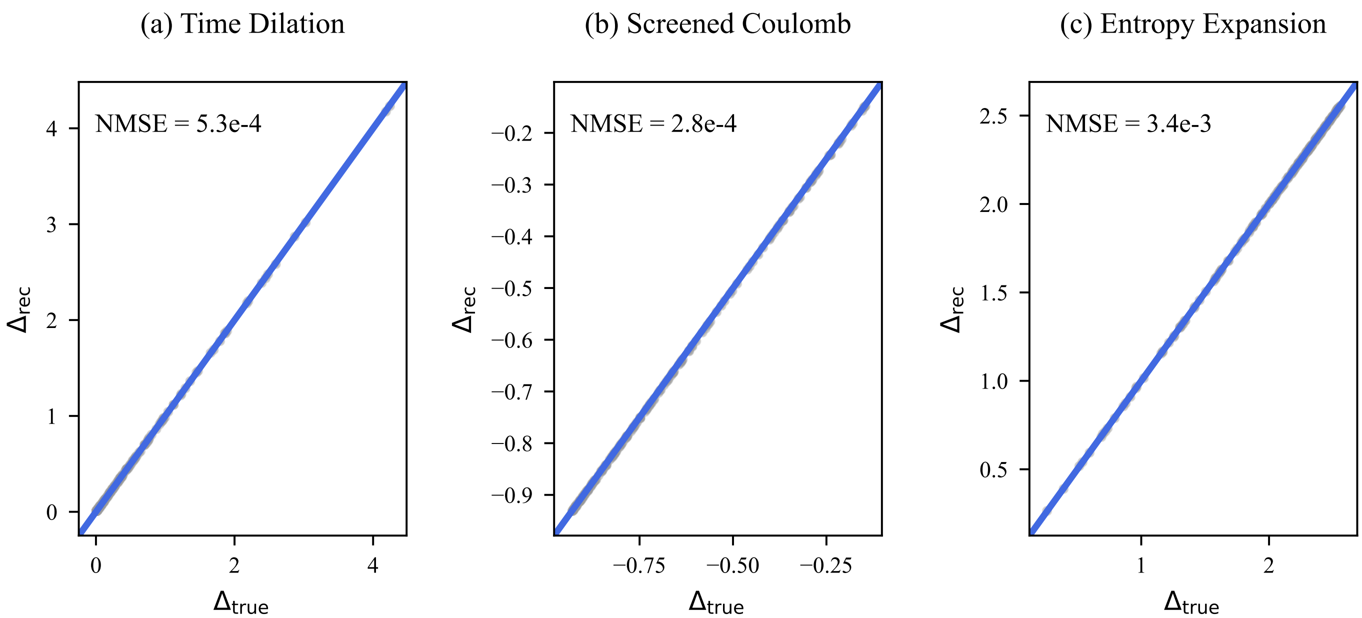

Figure 3: Parity plots: recovered correction (Δ_{rec}) vs. true correction (Δ_{true}) for all $N = 200$ data points. Blue solid line: 1:1 reference. Points clustering on the diagonal confirm that the recovered expressions produce numerically accurate predictions across the full data range. NMSE values are from the validation log (commit `f3cf7f3`).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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